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# GlycoGene CRISPR library for forward genetic screens of human glycosylation pathways

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## Abstract

CRISPR-Cas9 based forward genetic screens represent a powerful discovery platform to uncover genes regulating specific biological processes. This article describes the method to utilize a freely available "glycoGene-CRISPR library" to knock out any gene participating in human glycosylation in arbitrary cell types. The end product is a stable glycogene-CRISPR knockout cell library, where each cell contains one or more sgRNA, and lacks corresponding function. The cell library can be screened using various lectin/antibody reagents. It can also be applied in functional assays to establish glycan structure-glycogene-glycopathway relationships. This is a powerful Systems Glycobiology strategy to dissect glycosylation pathways and processes.

Basic Protocol 1: Scale up of the glycoGene CRISPR library and next-generation sequencing (NGS) library validation

Basic Protocol 2: Preparation of CRISPR lentivirus library and isogenic cell line stably expressing Cas9 nuclease

Basic Protocol 3: Establishing glycoGene CRISPR cell library, self-inactivation of Cas9 and NGS cell library validation

Basic Protocol 4: Enrichment of lectin binding or non-binding cells and related multiplex NGS data acquisition

Basic Protocol 5: Bioinformatics pathway analysis

### Keywords

CRISPR-Cas9; glycogene; glycosylation; lectin; pathway analysis

## INTRODUCTION:

Glycans are formed on the human/mammalian cell surface by the concerted action of various families of 'glycogenes' and corresponding 'glycoenzymes' including

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glycosyltransferases that assemble the glycans, glycosidases that break them down, epimerases, transporters, synthases, sulfotransferases etc (Neelamegham, Zhou, & Groth, 2022; Taniguchi et al., 2014). The final structure and function of carbohydrates formed on the cells, and the carbohydrate binding molecules (CBM) they recognize depends on the underlying cellular glycogene components and machinery. Additionally, the glycogenes participating in the formation of specific carbohydrate structures may vary among various cell types, as these are commonly isoenzymes. Thus, there is a critical need for experimental resources that can help elucidate the glycogene machinery that results in specific glycan types. The large-scale collection of such data can lead to systems-level understanding of glycosylation processes, and reveal new ways to fine tune related structures.

To address the above gap in knowledge, we recently developed a pooled 'glycoGene CRISPR' library targeting 347 glycosylation related genes or 'glycogenes' (Zhu, Groth, Kelkar, Zhou, & Neelamegham, 2021). Nine-to-eleven sgRNAs with high on-target editing efficiency and low off-target effects were designed for each target glycogene using a stringent machine learning algorithm described elsewhere (Doench et al., 2016). Here, the target genes we selected are involved in the biosynthesis of different types of glycoconjugates including N-glycans, O-glycans, glycolipids, glycosaminoglycans, GPI-anchored carbohydrates, O-fucose, O-GlcNAc, O-Glucose, O-Mannose and EGF domain specific O-linked GlcNAcTs (EOGTs). Gene editing in this system depends on the *Streptococcus pyogenes* Cas9 nuclease, and it occurs in a doxycycline dependent manner. Additionally, once gene editing is complete, it is possible to knockout out the Cas9 protein using the Self-inactivating CRISPR (SiC) system to result in a stable cell library (Kelkar et al., 2020). This stable cell library can be used in functional studies to establish glycogene-glycan relationships. Overall, the system described here is part of a growing family of tools available that use CRISPR based genome editing to dissect glycosylation-related pathways (Mondal et al., 2015; Stolfa et al., 2016).

This unit provides the detailed steps for the use of the 'glycoGene CRISPR' library resource (Figure 1). Basic protocol 1 describes the steps to scale-up the glycoene library and next generation sequencing (NGS) methods to quantify library representation. Basic protocol 2 describes the method to establish a stable Cas9 expressing isogenic cell line and scale-up for the CRISPR-library virus. Basic protocol 3 describes the steps required to establish a stable CRISPR-Cas9 knockout cell library, including the steps required to knockout Cas9 activity permanently in this system using the SiC system. Basic protocol 4 provides a detailed description of how to use the cell library to select for glycogenes regulating the formation of specific carbohydrate epitopes, including sample preparation for multiplex amplicon sequencing NGS methods. The final basic protocol 5 outlines the bioinformatics processing and analysis steps for data analysis of NGS data to establish glycogene-glycan relationships.

## BASIC PROTOCOL 1

**Scale up of the glycoGene CRISPR library and next-generation sequencing (NGS) library validation—**The glycoGene CRISPR library is obtained from Addgene. This library contains ~10 single-guide RNA (sgRNA) for each of the target glycogenes.

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The following section provides stepwise instructions on how to amplify the library without loss of guide representation, and how to quantify the representation of the resulting library. To this end, the library plasmid DNA is electroporated into commercial electrocompetent *E. coli* cells. pUC19 plasmid electroporation is performed in parallel to confirm the quality of the electrocompetent cells being used for library amplification. The pool of library transformed cells is scaled up and plasmid DNA is isolated to obtain an amplified glycoGene CRISPR library. A small amount of amplified library is used as template, and the sgRNA region is amplified from this using PCR. The size of the amplicon is confirmed using gel electrophoresis. The PCR reaction product is then gel purified, and a second PCR reaction is used to generate sequencing sample for NGS. This step introduces Illumina P5 and P7 handles, along with a sample index on the P7 side. Following sequencing, the distribution of sgRNA in the NGS data are analyzed to determine the representation of sgRNA in the scaled-up pooled library. A bell shaped sgRNA distribution is expected with few underrepresented sgRNA.

## Materials:

GlycoGene CRISPR pooled library DNA (10ng/µl, Addgene Pooled Library 140691, https://www.addgene.org/pooled-library/neelamegham-glycogene-crispr/)

LB Agar powder (Sigma L3027)

LB Broth powder (Sigma L3152)

Deionized and Milli-Q water

Carbenicillin Powder (Sigma C1389): Prepare a 100 mg/ml stock in water, aliquot and store at −20°C

NEB-10 Beta electrocompetent cells (NEB C3020K, or an equivalent strain from another vendor)

pUC19 plasmid (optional; included with the NEB-10 Beta electrocompetent cells)

Nuclease-free water (Invitrogen 10977015)

NucleoBond Xtra Midi (Macherey-Nagel, 740410–50 or equivalent)

Nucleospin Gel and PCR Clean-up kit (Macherey-Nagel 740–609)

NEBNext Ultra II Q5 Master Mix (NEB M0544)

1 M Tris, pH 8.0 (Invitrogen™ AM9856)

0.5 M EDTA (Invitrogen™ AM9261)

1X TE buffer (Please see recipe)

Agarose (National Diagnostics EC-202–500)

50X TAE Buffer (National Diagnostics EC-872, also see recipe)

Ethidium Bromide solution 10mg/ml (Invitrogen 15585011)

Gel loading Dye, 6X (New England Biolabs B7024S)

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GeneRuler Low Range DNA Ladder, Ready to use (Thermo Scientific FERSM1193)

Custom PCR primers (from any vendor such as Invitrogen, IDT, Eurofins Genomics etc.):

U6 TetO NGSlib Fwd (PAGE purified):
GACGCTCTTCCGATCTTTATATATCCCTATCAGTGATAGAGACACCG

NGSlib Rev (Desalted):
GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTCTTTAGTTTGTATGTCTG TTGCTATTATGTCTACTATTCTTTCC

P5 NGS Fwd (Desalted):
AATGATACGGCGACCACCGAGATCTGACGCTCTTCCGATCT

P7 Index NXXX Rev (Desalted):
CAAGCAGAAGACGGCATACGAGAT*NNNNNNNN*GTGACTGGAGTTCAGACG TGTGCTCTTCCGATCT

Analytical balance (e.g. Mettler Toledo ML203T; for a weight range of 1 mg to 220g)

245 mm Square Dish (Nunc™ 240845 or Corning™ 431111)

100 mm circular Petri dish (VWR 25384–342)

Glass bottles of various sizes

Autoclave oven

Cell spreader (Fisherbrand™ 14-665-230)

Cell scraper (Falcon™ 353085)

Electroporation cuvettes (1 mm gap, Bio-Rad 165–2083 or Fisherbrand™ FB101)

Bio-Rad MicroPulser (Bio-Rad 165–2100 or equivalent)

14 ml round bottom tube (Falcon™ 352059)

Shaker incubator

Incubator

NanoDrop™ (Thermo Scientific™ ND-ONE-W)

Thermocycler

Erlenmeyer flasks

Glass beakers of various sizes

Microwave oven

Beaker/flask tongs

Horizontal gel electrophoresis equipment with power supply

UV transilluminator

Plastic gel cutter razor blade (VWR 10048–876)

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Gel imaging and documentation system (e.g. Bio-Rad ChemiDoc)

## Amplification of Glycogene CRISPR Library

### One day before the library amplification

1. Weigh out 10 g LB agar powder. Dissolve in 200 ml deionized water and make up the volume to 250 ml.
2. Weigh out 6.25 g LB broth powder and prepare 250 ml of LB liquid medium

   Please check the bottle labels for the appropriate quantity for 250 ml medium, in case of using a different LB broth or LB agar brand or catalog number.
3. Sterilize both media by autoclaving (121°C, 15 min).
4. For LB agar medium only: After cooling to about 55°C, add 250 µl of 100 mg/ml Carbenicillin stock to obtain a final concentration of 100 µg/ml. Immediately pour ~150 ml into one 245 mm square dish and ~15 ml each in four, 100 mm circular Petri dishes. Allow to solidify completely and store at 4°C
5. Store the LB liquid medium at room temperature. Add carbenicillin to a final concentration of 100 µg/ml at the time of use.

### On the day of library amplification

6. Prewarm the outgrowth medium (provided with NEB-10-beta cells) and the LB agar plates at 37°C for at least 1 hour.
7. Place 1 µl of pooled library DNA (10ng total) in a microcentrifuge tube on ice.
8. Dilute the pUC19 control from the manufacturer 1:5 with sterile nuclease-free water to obtain a plasmid concentration of 10pg/µl and place 1 µl in another tube on ice. This is the positive control.
9. Place two electroporation cuvettes on ice.
10. Thaw a vial of NEB-10 Beta electrocompetent cells on ice and mix thoroughly by flicking gently.
11. Add 25 µl of the cells to the tube containing library DNA and mix by gently pipetting up and down 2–3 times. Incubate on ice for 1 min.
12. Transfer the mixture to the bottom of pre-chilled cuvette without introducing any bubbles and tap gently on the bench to spread the liquid on the bottom. Carefully wipe the outside of cuvette to remove any water or ice.
13. Electroporate following manufacturer's instructions for *E. coli* cells. (For Bio-Rad MicroPulser, use setting "Ec1" for *E. Coli* and 1mm gap cuvette.)

    Typically, time constant should be between 4.8 and 5.1 milliseconds.
14. IMMEDIATELY, add 975 µl prewarmed outgrowth medium to the cuvette, gently mix by pipetting up and down, and transfer to a 14 ml round bottom tube.

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Prompt addition of the outgrowth medium immediately after electroporation is critical for an efficient recovery of transformed cells. Any delays can reduce the transformation efficiency.

15. Repeat steps 7–14 for the pUC19 control plasmid.

16. Shake vigorously at 250 rpm at 37 °C for 1 hour.

17. Follow the separate guidelines below for the pUC 19 positive control and the glycoGene CRISPR library transformed cells.

## Plating to check the transformation efficiency (CRISPR library and positive control)

18. Serially dilute the library transformed cells as follows: first dilute 10 µl cells with 990 µl LB medium (100X) dilution. Mix 40 µl of the diluted cells with 960 µl of LB medium to get a 2500X final dilution. Use a cell spreader to plate 50 µl and 100 µl into two round Petri dishes containing LB agar medium with carbenicillin. Incubate overnight at 37 °C.

19. Next day, calculate the colony forming units (cfu) by the following formula: (#colonies x dilution factor x total volume of cells)/plated volume. Divide the cfu by the number of guides (3974 in this library) to get the approximate fold representation of each guide in the amplified library.

    A guide representation of ≥100 indicates a faithful replication of the library.

20. Dilute 10 µl of pUC19 transformed cells in 990 µl LB medium and mix well (100X dilution). Plate 50 µl and 100 µl into two round Petri dishes containing LB agar medium with carbenicillin. Incubate at 37 °C overnight and count the colonies next day.

    Typically, 50 µl plating volume should give about 25–50 colonies to indicate good electroporation efficiency ($5 \times 10^9 – 1 \times 10^{10}$/µg pUC19) of the competent cells.

    If a lower colony count is obtained, it is recommended to start with a new, fresh batch of commercial electrocompetent cells.

## Plating and culture of glycoGene CRISPR library transformed cells for library amplification

21. Spread uniformly the remaining 990 ml library transformed cells on the large LB agar plate, containing carbenicillin, and incubate overnight at 37°C.

22. Next day, you should observe a lawn of bacterial colonies on the plate. Add 10 ml LB + Carbenicillin medium on the agar surface and using a cell scraper, gently scrape the colonies off into the liquid. Repeat 2 more times and retrieve all the bacterial growth into a 30 ml pool in a 500 ml or 1 liter bacterial culture flask.

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23. Increase the volume to 200 ml with LB + Carbenicillin medium and grow overnight at 250 rpm, 37°C. The volume of growth medium can be increased if a bigger library plasmid yield is desired.

24. Next day, pellet the bacteria at 4500–6000 × *g* for 10–15 min. and purify the plasmid using NucleoBond Xtra Midi or a kit of your choice and following manufacturer's instructions. Dissolve the DNA in nuclease-free water.

25. Measure the DNA concentration using NanoDrop™ or an equivalent instrument. Store the library DNA at −20°C.

    It is recommended to have library DNA concentration of ≥ 1μg/μl and a 260/280 ratio of 1.8 or above to proceed with further steps.

## PCR and NGS of glycoGene CRISPR Library for checking guide representation

26. Dilute the plasmid library template to 50 ng/μl concentration in nuclease-free water.

27. Dissolve the primers (U6 TetO NGSlib Fwd, NGSlib Rev, P5 NGS Fwd, P7 Index NXXX Rev) to get 100 μM concentration stocks by adding 10 μl nuclease-free water or 1X TE buffer per nanomole of primer oligo.

28. From the 100 μM primer stocks above, prepare 1:10 dilution in nuclease-free water to get 10 μM primers to be used in the PCR reactions below.

29. PCR 1: Set up the following PCR reaction. Prepare the samples and perform PCR as detailed below.

    PCR 1 reaction setup (on ice):

| | |
|---|---|
| Nuclease free water | 14 μl |
| Template | 1 μl (50 ng) |
| 10 μM U6 TetO NGSlib Fwd | 5 μl |
| 10 μM NGSlib Rev | 5 μl |
| NEBNext Ultra II Q5 | 25 μl |

    PCR 1 parameters:

| | |
|---|---|
| Initial Denaturation | 98 °C, 30 sec |
| 10 cycles of: | |
| \|Denaturation | 98 °C, 10 sec\| |
| \|Annealing/Extension | 67 °C, 45 sec\| |
| Final Extension | 65 °C, 5 min |
| Hold | 4 °C |

30. Prepare 2% agarose gel by melting 2g agarose per 100 ml of 1X TAE buffer, in a glass flask or beaker, using a microwave.

31. While melting agarose, use medium power setting and keep an eye on the solution since it is susceptible to boiling over with excessive heat. You may need

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to intermittently take the container out and swirl the solution gently to allow complete dissolution of agarose and to prevent boil over.

Use extreme caution while handling hot agarose solution since it can boil eruptively and can cause serious burn injuries. Use a beaker/flask tong to handle the hot container.

**32.** Allow the agarose to cool down to about 60°C. Add 2 µl Ethidium bromide/100 ml of gel and mix homogenously by swirling.

Ethidium bromide is a known carcinogen, always wear gloves while handling it. If ethidium bromide is not mixed completely, bright non-uniform fluorescent patches will be seen while examining the gel under UV transilluminator.

**33.** Pour the agarose into a gel casting tray with a well comb already positioned inside. Allow the gel to solidify for 15–20 min.

Avoid pouring bubbles in the casting tray and if that happens, gently move the bubbles to the bottom of the tray with a clean pipette tip.

**34.** Mix 1 µl of the PCR product with 4 µl nuclease free water and 1 µl of 6X gel loading dye.

**35.** Load in one well of a 2% agarose gel. In another well, load 2–4 µl GeneRuler, low range DNA ladder (1 µl per mm width of the well comb).

**36.** Run the gel in 1X TAE running buffer for 20–30 min and examine on a UV transilluminator or using a ChemiDoc gel documentation instrument to confirm presence of 307 bp PCR1 product (Fig. 2A).

Wear gloves while handling the gel as ethidium bromide is a known carcinogen. Also wear UV safe goggles while examining gel under UV light to protect eyes from UV radiation damage.

**37.** If a single band at 307 bp is present, purify the reaction by loading on a column of Nucleospin Gel and PCR Clean-up kit and following the purification protocol included with the kit. Proceed to step 39.

**38.** If a strong 307 bp band with few other minor bands is present, load the entire reaction on a 1.8% preparative agarose gel prepared with wide comb. After electrophoresis in 1X TAE buffer at 100V for 20 min, excise the band at the 307 bp position. Follow the gel purification protocol using the same kit as in the step above.

**39.** Elute the column- or gel-purified PCR product in 15 µl elution buffer and proceed to PCR 2 reaction below.

**40.** PCR 2: Use the entire eluate from the step above, and perform a second PCR reaction for NGS sample preparation.

Use unique “P7 Index NXXX rev” primer for each reaction if processing multiple samples.

PCR 2 reaction setup (on ice):

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| | |
|---|---|
| PCR 1 Template (from above) | 15 µl |
| 10 µM P5 NGS Fwd | 5 µl |
| 10 µM P7 Index NXXX Rev | 5 µl |
| NEBNext Ultra II Q5 | 25 µl |

PCR 2 parameters:

| | |
|---|---|
| Initial Denaturation | 98 °C, 30 sec |
| 5 cycles of: | |
| \|Denaturation | 98 °C, 10 sec\| |
| \|Annealing/Extension | 65 °C, 45 sec\| |
| Final Extension | 65 °C, 5 min |
| Hold | 4 °C |

**41.** Follow the steps described above for agarose gel electrophoresis to check 1 µl of the PCR product on a 2% agarose gel. An amplicon of 364 bp should be observed in the PCR2 reaction (Fig. 2B).

Additional minor bands may be present besides the 364 bp product. Therefore, it is recommended to gel purify the PCR2 reaction.

**42.** Load the remaining PCR2 reaction on a 1.8% agarose gel. Excise the 364 bp DNA band and gel-purify using Nucleospin Gel and PCR Clean-up kit.

**43.** Elute the product in 15 µl elution buffer.

**44.** This is the NGS sample for the glycoGene CRISPR library.

**45.** Use the primer U6 TetO NGSlib Fwd as Read 1 primer for a 30 cycle NGS read on Illumina MiSeq platform (or similar) to evaluate guide representation in the amplified library. Although 20 cycles is sufficient to obtain necessary data, 30 cycles is necessary in order to obtain the NGS Quality Score (QS) metric. Allow ~1000 reads per sgRNA in order to obtain sufficient representation.

**46.** The expected sgRNA distribution in glycoGene CRISPR library is presented in Fig. 2C, based on previous studies. Detailed method to generate this figure is part of Basic Protocol 5.

## BASIC PROTOCOL 2

**Preparation of CRISPR lentivirus library and isogenic cell line stably expressing Cas9 nuclease—**The plasmid library pool obtained from Basic Protocol 1 is used to prepare a lentivirus library of glycoGene CRISPR sgRNAs in Lenti-X 293T cells. Any cell line of interest for intended use in a GlycoCRISPR editing experiment needs to be modified for stable expression of Cas9 nuclease. The following section provides a stepwise procedure for producing lentivirus particles. These steps will be used for both preparing the glycoGene CRISPR lentivirus library as well as for making lentivirus for stable transduction and expression of Cas9 nuclease in a cell line of interest and establish isogenic clones thereof. Alternatively, it is possible to commercially obtain a stable Cas9 expressing cell line, if available, for use in this experiment. A biosafety level 2 (BSL2) hood is required for

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handling cells and lentivirus preparation. Please familiarize yourself with your institution's safety regulations and strictly follow the guidelines throughout this procedure.

### Materials:

Lenti-X™ 293T Cells (Takara Bio 632180)

SiC-V1-Scr (Addgene 133042; Lentivirus vector for stable expression of spCas9 and dTomato reporter; Size 13114 bp)

GlycoGene CRISPR Library plasmid (Scaled up in Basic Protocol 1; Size 8536 bp)

psPAX2 (Addgene 12260; lentiviral packaging plasmid; Size 10709 bp)

pMD2.G (Addgene 12259; VSV-G envelope expressing plasmid for lentivirus production; Size 5822 bp)

DMEM-complete (see recipe)

DMEM high glucose, pyruvate (Gibco™ 11995065)

Fetal Bovine Serum (Gibco™, 26140079)

Antibiotic-Antimycotic (100X; Gibco™ 15240062)

GlutaMAX™ Supplement (100X; Gibco™ 35050061)

Trypsin-EDTA (0.25%), phenol red (Gibco™ 25200072)

PBS, pH 7.4 (10X; Gibco™ 70011069)

OPTI-MEM™ Reduced Serum Medium, GlutaMAX™ Supplement (Gibco™ 51985034)

2X HBS (see recipe)

2M Calcium Chloride (Sigma C5670, 11.1 g $CaCl_2$-anhydrous in 50 ml final volume of MilliQ water, filter sterilize, store at 4°C)

Note: it is important to use anhydrous $CaCl_2$ for this preparation

50 mM Chloroquine diphosphate (Sigma, C6628, 130 mg in 5 ml final volume of MilliQ water, filter sterilize, store at 4°C)

500 mM Sodium butyrate (Sigma 303410, 27.7 g in 100 ml final volume of MilliQ water, filter sterilize, store at 4°C)

Polybrene (Sigma H9268, prepare a 1000X stock at the concentration of 8mg/ml in milliQ water, filter sterilize, store at 4°C)

IMDM-complete (see recipe)

1X HEPES buffer (see recipe)

T-175 Tissue-culture flasks (Greiner Bio-one 660175)

150 mm round Tissue-culture dish (Corning 353025)

Tissue culture plates (96 wells, 48 wells, 24 wells, 12 wells and 6 wells)

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Sterile serological pipettes for tissue culture (Greiner Bio-one 606180, 607180, 760180)

15 and 50 ml conical tubes

Micropipettes (e.g. Eppendorf 2231300006)

Micropipette tips

0.45 μm PES syringe filters

0.45 μm PES bottle top filters

0.2 μm PES syringe filters

Ultracentrifuge tubes (refer to the model of your ultracentrifuge for correct tubes)

NucleoBond Xtra Midi (Macherey-Nagel, 740410–50)

1.7 ml Microcentrifuge tubes

Flow cytometer tubes (Falcon™ 352052)

Strainer caps with 35 μm mesh, for flow cytometer tubes (VWR 76449–662)

BSL2 Biosafety cabinet

Vacuum aspirator (in biosafety cabinet)

Water bath or dry bead bath at 37°C for warming up cell culture media

Cell culture incubator (Humidity controlled and $CO_2$ supplied)

Hemacytometer (Hausser Scientific™ 3100)

Tabletop centrifuge with hanging bucket rotors, for 15 and 50 ml conical tubes

Inverted Microscope

Fluorescence microscope

Ultracentrifuge

Microcentrifuge

Liquid nitrogen storage tank for cell lines

Flow Cytometer

## Lentivirus particle preparation

1. Maintain Lenti-X™ 293T cells in DMEM medium containing 10% FBS, 1X Antibiotic-Antimycotic and 1X Glutamax™ Supplement (DMEM-complete).
2. Pass the cells at 80–90% confluency.

   Discard if 100% confluent or overgrown, and thaw a new vial of frozen cells.

   Use early passage cells for creating lentivirus.

   Always pre-warm the cell culture medium to 37°C before adding it to the cells.

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3. For large scale production of lentivirus, grow cells in T-175 flasks to 80% density. Pass one such flask into 3–4, 150 mm round cell culture dish.

   Depending on desired scale of virus production, have enough number of T-175 flasks ready one day before the transfection protocol. Recommended number is: 5–10 150 mm culture dishes for SiC-V1-scr lentivirus and at least 2–4 such dishes for preparing glycoCRISPR library virus.

4. Aspirate the growth medium from the flask and add 10 ml of 1X PBS gently on the side wall of the flask without disturbing the cell layer. Manually rock the flask gently to rinse away any remaining growth medium and remove the PBS by aspiration.

5. Add 5 ml Trypsin-EDTA solution and rock gently to spread the trypsin uniformly on the cell layer. Move the flask to incubator for 1 min. Check under a light microscope for rounding of the cells and gently rock and tap the flask to detach the cells into trypsin.

6. Add 10 ml of DMEM-complete (FBS inactivates the Trypsin) and collect the detached cells by washing the medium along the bottom of the flask using a serological pipet. Remove the cells into a 50 ml conical tube.

7. Centrifuge the cell suspension at 250 × g for 5 min, preferably in a hanging bucket rotor. Aspirate the supernatant carefully without disturbing the cell pellet.

8. Add 1 ml DMEM-complete to the pellet using a 1 ml micropipette and gently pipet up and down 8–10 times to make a homogenous cell suspension.

9. Seed the cells into 3–4, 15 cm cell culture dishes containing 20 ml DMEM-complete. Gently rock the dish back and forth and side to side to uniformly distribute the cells.

   Do not rotate the dish to distribute the cells as this may cause most cells to seed in the middle and create non-uniform cell distributions.

10. Allow the cells to grow in a humidity controlled, 37°C incubator with 5% $CO_2$ overnight.

11. Next day, check the cell density under the microscope to make sure that it is between 50–60%.

    Transfection efficiency with calcium phosphate is best at 50–60 % cell density.

12. Aspirate the growth medium and add 20 ml of prewarmed DMEM-complete to each dish. Return the dishes to the incubator.

    It is best to transfect cells within 1–2 hours after the medium change.

13. For transfections, mix the Transfer plasmid (e.g. SiC-V1-scr; 13114 bp or GlycoGene CRISPR library plasmid; 8536 bp), the Packaging plasmid (psPAX2, 10710 bp) and the Envelope plasmid (pMD2.G, 5822 bp) at 2:2:1 ratio respectively based on their molecular size in bp. A total of 45–55 μg of the plasmid mix is used for each 150 mm dish.

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The plasmids used for lentivirus preparation should be of high purity.

It may also help to bring all the plasmids to 1 μg/μl concentration for ease of calculation.

For making lentivirus for Cas9 nuclease expression using SiC-V1-scr plasmid (Fig. 3A, top), mix 26228 ng (2 X 13114 bp) of transfer plasmid, 21418 ng (2 X 10709 bp) of psPAX2 and 5822 ng (1X 5822 bp) of pMD2.G plasmid.

For making lentivirus using GlycoGene CRISPR library plasmid (Fig. 3A, bottom), use 17072 ng (2 X 8536) of transfer plasmid with 21418 ng of psPAX1 and 5822 ng of pMD2.G.

**14.** For each 150 mm dish, prepare 2 ml transfection mix (volume of the transfection mix should be 10% of the volume of growth medium in the dish).

**15.** To the plasmid DNA mix prepared in previous step, add cell culture grade water to bring the volume to 875 μl. Add 125 μl of 2M $CaCl_2$ drop-wise and mix gently. Keep at room temperature for 5 min.

**16.** Add 1 ml of 2X HBS solution dropwise and then rapidly mix the solution by vigorous bubbling with the pipette.

**17.** Incubate the transfection mix at room temperature for 15–20 min. The mixture should turn visibly cloudy at this point.

**18.** Five minutes before adding the transfection mix to the plate, add 10 μl of 50 mM chloroquine to each 150 mm dish (5 μl of 50 mM chloroquine for each 10 ml medium for a final concentration of 25 μM).

**19.** Add 2 ml transfection medium to each dish by uniformly distributing in gentle drops all over the surface of the growth medium.

**20.** Gently rock the dish to mix and return to the incubator.

**21.** After 6–8 h post transfection, aspirate the medium from each dish and add 20 ml of pre-warmed Opti-MEM. Incubate the dish for 18–20 hours.

**22.** About 24 hours post transfection, check the cells under a fluorescent microscope to check the transfection efficiency. The SiC-V1-scr plasmid has a dTomato reporter (Ex:554nm, Em:581nm) and the glycoGene CRISPR Library plasmid has a BFP reporter. Minimally 60% (typically >80%) of the cells should be positive for the fluorescence reporter. If the transfection efficiency is lower, please refer to troubleshooting step and repeat the experiment.

**23.** Collect the Batch I of lentivirus in a 50 ml conical tube. Add 15 ml of prewarmed Opti-MEM supplemented with 300 μl of 500 mM Sodium butyrate (10 mM final concentration of Sodium butyrate).

**24.** Return to incubator for 18–20 hours.

**25.** Collect the Batch II of lentivirus and pool with Batch I in the 50 ml Conical tube.

**26.** Spin at 1000 × g for 2 min to remove the cell debris.

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27. Gently filter the supernatant through 0.45 μm PES filters (syringe filters or bottle top filters may be used depending on the volume of the lentivirus pool being processed).

28. Transfer the filtered lentivirus pool to appropriate ultracentrifuge tubes. If using a fixed angle rotor, mark the side of the tube facing the rotor's outer wall and spin at 50,000 × g for 2 hours at 4 °C.

29. Carefully remove the tubes from the rotor. A glassy pellet may be observed on the marked side of the tube.

30. Carefully decant the supernatant in a waste container marked for lentivirus waste collection.

31. Line the cell culture hood surface with aluminum foil topped with kimwipes and invert the tubes for 5–10 min to drain the supernatant.

32. While keeping the tube inverted, and using a micropipette tip, try to remove any supernatant that may have accumulated along the rim of the tube.

33. Add 100 μl of Opti-MEM™ to the pellet, set a micropipette at 60 μl and resuspend the pellet without introducing any bubbles.

    Most of the times, total volume of resuspended virus will be more than 100 μl because some supernatant invariably remains from the previous step. Depending on the volume recovered, we will get 200–300X concentrated virus.

34. If the same lentivirus is processed in multiple tubes, pool all the dissolved pellets into a 1.7 ml microcentrifuge tube. Centrifuge at the highest speed of the microcentrifuge for 1 min.

35. Aliquot small volumes into multiple tubes and store at −80 °C until use.

### Establishing stable Cas9 cell line using SiC-V1-scr lentivirus transduction

36. Use early passage cells to establish the stable Cas9 expressing cell line.

    This example uses HL60 cells (Fig. 3B i). These cells grow in suspension and do not require trypsinization.

37. Transfer the cells from a T175 flask to a 50 ml conical tube and centrifuge at 250 × g for 5 min.

38. Aspirate the growth medium and resuspend in 1 ml IMDM-complete.

39. Count cells using a hemacytometer and bring the cell density to $10 \times 10^6$/ml.

40. Dispense 20 μl cells (200,000) each in 4 microcentrifuge tubes.

41. Dilute the 1000X polybrene stock 1:100 in IMDM-complete to get a 10X stock (80 μg/ml polybrene).

42. Add 10 μl of 10X polybrene stock to each of the 4 tubes.

43. Add 0, 5, 25 and 50 μl of lentivirus prepared with SiC-V1-scr transfer vector (as described in the section above).

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**44.** Add appropriate amount of IMDM-complete to bring the total volume to 100 µl in each tube. Incubate at room temperature for 10 min.

**45.** Transfer to 4 wells of a 12 well plate each containing 600 µl of IMDM-complete with 8 µg/ml polybrene so the cells are covered with a thin layer of medium. Keep for 2 h in incubator.

**46.** Add 300 µl growth medium containing 8 µg/ml polybrene to each well and continue incubation for 24 h.

**47.** Next day, replace the culture medium with 1ml of fresh IMDM-complete.

**48.** Check the cells under the fluorescence microscope for dTomato expression which is a surrogate reporter for Cas9 nuclease expression in the cells.

Fluorescence signal may be initially weak but will increase by 48 to 72 hours.

**49.** When the cells are confluent (~48–72 h) remove 100 µl cells from the culture well into a microcentrifuge tube.

**50.** Pellet the cells at 250 × g for 5 min.

**51.** Aspirate the supernatant and resuspend the cell pellet in 1X PBS.

**52.** Analyze the cell population by flow cytometry using appropriate wavelength for dTomato.

**53.** For the wells showing high level expression of dTomato (Fig. 3B ii), pass the cells into 6 well plate and expand further from there gradually to obtain a T-175 flask. This may take 7–10 days depending on the cell type.

This polyclonal pool of cells may contain single to multiple copies of randomly integrated Cas9-dTomato cassettes in the genome.

**54.** Freeze a few vials of the polyclonal pool of stable Cas9 cells and store under liquid $N_2$.

**55.** Prepare cells for bulk sorting by pelleting the cell suspension and resuspending in 1X HEPES buffer at a density of $10 \times 10^6$ cells/ml.

**56.** Filter the cell suspension through a 35 µm strainer cap into a flow cytometer tube, just before sorting.

This step breaks down any cell clumps to produce a single cell suspension for flow cytometry.

**57.** In parallel, also prepare untransduced, control HL60 cells in a similar manner to serve as control and aid in creating gates for sorting.

**58.** Prepare a 15 ml collection tube with 1 ml IMDM-complete medium.

**59.** For flow cytometer sorting, set the gate for the top 10–20% dTomato expressing cells and bulk sort ~ 100,000 cells into the prepared collection tube.

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**60.** Spin down the tube at 250 × g for 5 min in a hanging bucket rotor and carefully aspirate most supernatant leaving a small volume in the tube so that the cell pellet is not disturbed or aspirated.

**61.** Resuspend the sorted cell pellet in 1 ml IMDM-complete medium and transfer to a well of 24 well plate.

**62.** Keep moving the cells to one size bigger container as they become confluent until you have a T175 flasks of confluent cells.

Do not discard any cells while passing.

**63.** Freeze a few tubes of the bulk sorted Cas9+ cells if you need to go back to these later.

**64.** Next, establish isogenic clones of high Cas9 expressing cells by sorting bulk sorted cells into single cells per well of a 96 well plate by FACS or serial dilution into 96 well plate.

**65.** For single cell sorting, prepare the cells as in the steps above for bulk sorting. Also prepare a 96 well plate by putting 100 µl medium (IMDM-complete and conditioned IMDM-complete mixed in 1:1 ratio), in each well. See the following step for preparing the conditioned medium.

**66.** Some cell lines may not grow very well when single sorted. To support their growth, sorting and initial culture is done in conditioned medium. Conditioned medium is prepared by seeding the same cell type in a T-75 or T-175 flask such that it will grow to 50–60% confluency by next day. Next day, the medium is collected, spun down at 250 × g for 5 min to remove any cell debris and filtered through a 0.2 µm PES filter. This medium is then used to support the growth of single sorted cells. This step is particularly important when single cell sorting HL-60 cells, as the conditioned medium contains cell-secreted factors that enhance recovery.

**67.** It may take 1–2 weeks for single cells to multiply into a population and become visible in suspension (or as colonies if adherent to the plate). Monitor the plates every 2–3 days and supplement with 25–50 µl of growth medium every week or so.

**68.** Check the plate under the microscope upon achieving significant confluence/density and transfer to 24 well plates, as the cell density increases. Continue to expand the isogenic clones and freeze down cells when achieving culture in T25 flask.

**69.** Evaluate a fraction of the cells for the expression of dTomato by flow cytometry and microscopy to ensure high levels of Cas9 (Fig. 3B ii). Compare the glycosylation status of these newly sorted cells with respect to wild type cells using lectin profiling as necessary. In our experience, we have observed that the procedure does not alter cell glycosylation status, as judged based on lectin profiling and also selectin binding function (see Basic Protocol 4 below).

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## BASIC PROTOCOL 3

### Establishing glycoGene CRISPR cell library, self-inactivation of Cas9 and cell library validation by NGS

**Introduction:** This protocol outlines the steps to create a library of glycoGene CRISPR edited cells. There are two controls built into this system to limit the off-target editing effects of CRISPR-Cas9. First, the expression of sgRNAs in the glycoGene CRISPR library is doxycycline-inducible and this allows the user a control over the time window of gene editing. This also allows flexibility in choosing the length over which editing is allowed in the experiment, since some sgRNAs may be more potent than others, and also some genes may be more accessible for editing than others depending on their chromatin structure. After the desired period of controlled editing, doxycycline is removed from the medium to stop editing. Second, to prevent sgRNA non-specific or off-target editing after preparation of stable cell library, a potent sgRNA targeting the Cas9 gene is electroporated into the library cells. This disables the functional activity of the Cas9 nuclease. Together, the above control protocols enable the construction of a longitudinally stable population of glycoGene CRISPR edited cells. The NGS of this population allows assessment of sgRNA representation in the cell library before functional screens are undertaken.

### Materials:

Cas9G7 sgRNA (Synthego custom order, CRISPRevolution sgRNA EZ kit-Cas9G7 UACGCCGGCUACAUUGACGG – Modified)

Neon™ Transfection System (Invitrogen™ MPK500)

Neon™ Transfection System 100 µl kit (Invitrogen™ MPK10025)

GlycoGene CRISPR Library Lentivirus (Prepared in Basic protocol 2)

Doxycycline (Sigma D9891, prepare a 5000X stock at 5 mg/ml in milliQ water, Filter sterilize, store at −80°C)

PureLink™ Genomic DNA mini kit (Invitrogen K182001)

Additional reagent and equipment for tissue culture and transduction (from Basic Protocol 2)

### Titration of GlycoGene CRISPR library by transduction and flow cytometry

1. Prepare glycoGene CRISPR library virus using the lentivirus preparation protocol outlined above (Basic Protocol 2: Virus particle preparation)
2. Perform the titration on the cell line which is intended for the CRISPR experiments as the cell lines vary in transduction efficiency. The following procedure outlines a titration experiment using HEK 293T cells.
3. Trypsinize HEK 293T cells and count. Dilute to $2 \times 10^6$/ml. Dispense 100,000 HEK 293T cells in five tubes and make up the volume to 100 µl with growth medium.

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4. Dilute the 1000X polybrene solution 10-fold to get a 100X stock (at 0.8 mg/ml). Add 1 µl of the 100X polybrene to each tube of cells to get a final concentration of 8 µg/ml.

5. Prepare a 1:10 dilution of the glycoGene CRISPR Library lentivirus by mixing 1 µl of virus with 9 µl of growth medium. Add 0, 0.1 (1 µl of 1:10 dilution), 0.25 (2.5 µl of 1:10 dilution), 1 (undiluted) or 2.5 µl (undiluted) virus and incubate at room temperature for 10 min.

6. Transfer the cells to the wells of 24 well plate and add 150 µl growth medium containing 8 µg/ml polybrene so the cells are covered with a thin layer of medium. Keep for 2 hr in incubator.

7. After 2 h, add 250 µl more growth medium containing 8 µg/ml polybrene to each well and keep in the incubator overnight.

8. Next day aspirate the growth medium from each well to remove virus, and add fresh DMEM-complete medium. Check under microscope for expression of fluorescent marker. Some fluorescence will be seen.

9. Grow cells for 48h to allow greater fluorescence signal development and assay by flow cytometry as follows.

10. Aspirate growth medium from the wells and add 200 µl 1X PBS. Gently swirl the plate and aspirate the PBS.

11. Add 100 µl Trypsin-EDTA to each well. Return the plate to the incubator for 1 min.

12. Gently tap the plate and check under the microscope for complete detachment of the cells.

13. Add 200 µl DMEM-complete and pipet up and down to resuspend the cells. Transfer to a 1.5 ml microcentrifuge tube. Spin down at 250 × g for 5 min and aspirate the supernatant without disturbing the cell pellet.

14. Add 200 µl of 1X PBS and resuspend to make a homogenous solution. Read the samples using a flow cytometer with appropriate wavelength for the dTomato and BFP reporters (Fig. 3B iii).

15. Use the well with <50% transduction for calculating the virus titer using following formula.

    Titer = (No. of cells seeded) * (% reporter positive cells/100) / (vol. of virus added in ml)

    Following is a calculation for 30% cells expressing the fluorescent marker in the well transduced with 0.25 µl virus:

    Titer = (100,000)*(30/100)/(0.00025 ml) = $1.2 \times 10^8$/ml

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### Creating glycoGene CRISPR library

**16.** The following protocol is given for HL60-Cas9+ isogenic clones established by transduction with SiC-V1-scr lentivirus (in Basic Protocol 2).

**17.** Using the transduction protocol outlined above, transduce the Cas9+ cells with the library virus at low multiplicity of infection (MOI) of 0.1 to 0.4 (i.e. 10–40% of transduced cells) so each cell receives ≤ 1 virus.

Typically, the number of cells transduced should be 500–2000 times the total number of sgRNA in the library.

Thus, for a library of 4000 sgRNAs, transduce 2–8 × $10^6$ cells at MOI of 0.1 to 0.4.

For example, if the estimated virus titer is 1.2 × $10^8$/ml, 1.66 µl and 6.66 µl virus will be required for transducing 2 × $10^6$ cells to obtain MOI of 0.1 and 0.4, respectively. Note that this can vary in cases where the transduction efficiency (i.e. viral tropism) of the titer cell system (HEK in above example) is different from the final cell where the CRISPR library is expressed (HL60 in this example). It is important to establish the virus titer in the cell line of interest.

**18.** Two-days post-transduction, check the cells under microscope for Blue Fluorescent Protein (BFP, Ex:402nm Em:457nm) expression. Also, read a small portion of the cells by flow cytometry to determine the fraction of transduced cells. If this falls within 10–40% range proceed with the next step (Fig. 3B iii).

**19.** Pass the remaining cells into 2–4 T-175 flasks and start doxycycline treatment at 1 µg/ml (typical for many cells) to induce expression of the library sgRNAs and initiate gene editing in the cells.

It is critical to add doxycycline to the culture medium to trigger the expression of sgRNA since it is under the control of a tetracycline inducible H1 promoter. Failure to add doxycycline will result in absence of or low levels of gene editing.

**20.** Four-days post-transduction, terminate the doxycycline treatment and electroporate 1.5–2.5 × $10^6$ cells with 10 µg Cas9G7 using a 100 µl tip (Neon™ Transfection System, Invitrogen™).

**21.** The electroporation conditions are as follows (for HL-60): 1600 V, 10 ms, 3 pulses. Following electroporation, transfer the cells into six-well plates containing 3 mL of IMDM-complete.

Note that the electroporation condition varies with cell type and the user is referred to the Neon Protocol library. Additionally, electroporation may result in some cell death.

**22.** Examine a small fraction of the electroporated cells by flow cytometry on Day 8. Due to editing of the Cas9-dTomato cassette by electroporation of Cas9 sgRNA, some loss of dTomato reporter may be visible in a small population of cells. This negative dTomato population should progressively increase every 2–3 days

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as Cas9 knockout gene editing proceeds. Thus, a larger dTomato- population will emerge.

**23.** Maintain the cells by passing all the cells to bigger or multiple cell culture dishes for few passages after electroporation, to avoid loss of sgRNA representation.

It is a good practice to freeze cells in liquid nitrogen on different days during the procedure and label them appropriately in case one needs to go back to an intermediate step at a later date. It is also important to pass one half to one third of the library cells every time to maintain the guide representation in the library.

**24.** On day 12–15 post-transduction, bulk sort cells for BFP+ and dTomato- (Cas9-) population (Fig. 3B iv, arrow).

This is the self-inactivated CRISPR cell library.

**25.** Expand the library of cells for further functional screens (see Basic Protocol 4).

**26.** Additionally, analyze a portion of the CRISPR cell library around Day 20–25 to ensure that all the cells are BFP+ and dTomato-.

**27.** Prepare a vial of ~$5 \times 10^6$ cells for deep sequencing and also freeze a few vials of the library cells in liquid nitrogen.

### Deep sequencing of the GlycoGene CRISPR library cells

**28.** Prepare genomic DNA from ~$5 \times 10^6$ cells using PureLink™ Genomic DNA mini kit (or a similar kit from another vendor) using the kit protocol.

**29.** Measure the DNA concentration using NanoDrop.

**30.** Follow the PCR and NGS protocol described in the section “PCR and NGS of GlycoGene CRISPR Library for checking guide representation**”** (BASIC PROTOCOL 1). Set up three PCR1 reactions each using 1 μg of genomic DNA template and then proceed with subsequent steps. (Note: ~7μg genomic DNA content per $10^6$ cells, using 3 μg DNA will give us a representation of ~100 fold per sgRNA i.e $10^6$*3/7/#sgRNAs). Pool the reactions after PCR2.

**31.** Perform NGS using Illumina Sequencer as described in BASIC PROTOCOL 1, and analyze the NGS data using protocol in BASIC PROTOCOL 5. Typically allow ~1000 reads/sgRNA to obtain sufficient representation.

**32.** The above steps will result in the quantitation of the guide representation in the stable glycoGene CRISPR library pool. A bell-shaped curve (Fig. 4) indicating a uniform distribution and representation of guides should be seen, in the cell library.

## BASIC PROTOCOL 4

### Enrichment of lectin binding or non-binding cells and related multiplex NGS data acquisition

**Introduction:** Once a stable cell library consisting of the GlycoGene CRISPR edited cells is established, forward genetic screens are possible using both antibodies (Ab) that recognize

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carbohydrate related antigens and lectins that directly bind selected glycoconjugates. In such studies, the entire cell population is subjected to Ab/lectin screens that select for either a population that displays enhanced lectin/mAb binding (positive selection) or reduced binding (negative selection). In this protocol, such selection is performed using flow cytometry based sorting (FACS). Once a sub-population displaying selected phenotype is selected, it undergoes NGS based analysis to quantify sgRNA that are enriched in that population. Such analyses both confirms the function of gene based on prior biochemical knowledge, and reveals novel regulators of glycosylation. Since there are ~10 sgRNAs for each glycoGene in our library, enrichment of multiple sgRNA for any single gene helps corroborate the findings and confirm the absence off-target effects in the functional screen. A detailed example is provided below for P-selectin PSGL-1 binding negative-selection (Fig. 5A), but the same could be done for other lectins.

### Materials:

1X HEPES buffer (see recipe in Reagents and Solutions)

HEPES/Calcium/HSA (see recipe in Reagents and Solutions)

Human serum albumin (HSA), injectable from any source (e.g. Grifols)

Recombinant Human P-Selectin/CD62P Fc Chimera Protein (R&D systems 137-PS, reconstitute at 100 μg/ml in sterile PBS)

Alexa Fluor® 647 AffiniPure F(ab')$_2$; Fragment Goat Anti-Human IgG, Fcγ Fragment Specific (Jackson ImmunoResearch 109-606-098, 1.5 mg/ml)

Reagents and supplies from previous protocols

### Functional screen using GlycoGene CRISPR library cells

1. Pellet the HL60 library cells to remove the growth medium and wash with HEPES buffer twice. In parallel, also process HL60 Cas9+ cells (no sgRNA) in the same manner to serve as positive control.
2. Resuspend the cells in HEPES buffer containing 0.1% HSA and 1.5mM $CaCl_2$, and use a cell counter to count. Adjust the final cell density to $20 \times 10^6$ cells/ml and keep on ice.
3. The following screen is for the enrichment of edited cells that display reduced P-selectin binding (i.e. negative-selection).
4. Prepare 10X working stocks of P-selectin-Fc (dilute 1: 2 in HEPES/Calcium/HSA, 50 μg/ml), and Alexa-647 Anti-Human IgG (Dilute 1:100, 15 μg/ml)
5. Mix 12 μl each of 10X P-selectin-Fc and 10X Alexa-647 Anti-Human IgG with 96 μl of HEPES/HSA/$CaCl_2$, and incubate at room temperature for 10 min to allow complexation.
6. Mix 20 μl of the complex with 50 μl of HL60 Cas9+ cells (no sgRNA, $20 \times 10^6$/ml) and keep on ice for 15 min.

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7. Add 250 μl of 20 × $10^6$/ml HL60 library cells to the remaining 100 μl of the complex, and incubate on ice for 15 min.

8. Read the unlabeled HL60 Cas9+ cells and P-selectin bound HL60 Cas9+ cells (from step 6, washed once with 200 μl HEPES buffer) to set up the gates for BFP, dTomato and Alexa-647 signals to aid in subsequent sorting of the library cells with reduced P-selectin binding.

9. Add 1 ml of HEPES buffer to sample from step 7, perform a quick wash by spinning pellet and resuspending. Then sort for BFP+, dTomato-, Alexa 647 (P-selectin)-low cells (Fig. 5B).

   Ensure that the time between the wash step and sorting is kept to the minimum as this will minimize any P-selectin IgG unbinding.

10. At the end of the sort, spin down the sorted cells at 250 × g for 5 min in a hanging bucket rotor.

11. Carefully aspirate the supernatant leaving a little solution in the tube so that pellet is not disturbed

    Pellet may not be visible if there are only few thousand cells in the tube.

12. Resuspend the pellet in IMDM-complete and transfer to an appropriately sized cell culture dish depending on the number of cells recovered. Conditioned medium may be used to help cells recover better from the shock of sorting.

13. It may be required to perform 1–2 additional serial sorts to enrich the P-Selectin-low population of edited cells, in case the first sort does not yield a distinct population (Fig. 5B).

14. Once the cells are expanded, confirm loss of P-selectin binding using cytometry (Fig. 5C).

15. Prepare 2–3 vials of ~5 × $10^6$ cells for deep sequencing and also freeze a few vials of the sorted P-selectin-low cells in liquid nitrogen for later use as needed.

16. Also prepare 2–3 vials of ~5 × $10^6$ of the unsorted library cells to serve as a reference for NGS.

17. Prepare samples for deep sequencing by following steps in the last sections of Basic Protocol 1 and 3. Use 1–3 μg genomic DNA for PCR. Label each sample/replicate with a unique index primer (*P7 Index NXXX*) in PCR2.

18. Obtain ~1000 reads per sgRNA during the NGS run.

19. Data analysis is described in Basic Protocol 5.

## BASIC PROTOCOL 5

### Bioinformatics pathway analysis

**Introduction:** The CRISPR Cas9-based screens result in non-homologous end-joining (NHEJ) that causes gene insertions/deletions (indels) at the sgRNA target site and loss of

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corresponding function. Disrupting glycoenzymes using this approach should alter glycan synthesis, and thus should cause changes in lectin or antibody binding. Determining which gene(s) affects mAb/lectin binding in forward genetic screens requires statistical analysis. Here, we use the Model-based Analysis of Genome-wide CRISPR/Cas9 Knockout (MAGeCK) method to identify genes responsible for losses or gains in lectin binding as a result of a glycogene knockout. (Li, Xu et al. 2014). The steps involved in this process include: i) Installation of MAGeCK in the Linux environment. ii) Read counting and normalization for each of the sgRNA in each of the fastq experimental data or treatment files. iii) Computing $\log_2$ fold-change values of sgRNAs in sorted populations with respect to the starting unsorted population. and iv) Using α-Robust Rank Aggregation (RRA) to identify genes whose sgRNAs are consistently enriched/depleted in sorted cell populations. As an ancillary analysis, log-fold change plots are created to ensure consistency with the results generated from MAGeCK.

***Materials:*** Running MAGeCK is documented at the project's sourceforge page:

https://sourceforge.net/p/mageck/wiki/Home/. Its stepwise usage when analyzing the glycogene-CRISPR library is discussed below.

**Installation of virtualbox and miniconda—**The data analysis is best performed using Linux OS as that is supported by bioconda (https://bioconda.github.io/). MacOS instructions are similar. Bioconda does not support Windows. If working on Windows machine, install a Linux emulator/virtual machine like VirtualBox (www.virtualbox.org). This can be setup for a range of Linux distributions. The usage below is described using Ubuntu 20.04LTS. Follow this link for detailed stepwise installation of Ubuntu on VirtualBox: https://www.youtube.com/watch?v=x5MhydijWmc

1. Download miniconda from https://docs.conda.io/en/latest/miniconda.html. Choose Miniconda3 Linux 64-bit with Python 3.9 and download into "Downloads" folder. Open Linux terminal and install miniConda by stepwise implementing the commands listed in Table 1. The left column provides the command while right column the meaning:

2. Install mageck using miniconda by sequentially following the steps in Table 2.

### Obtaining count data and perform normalization

3. The following input files are needed for MAGeCK: **1)** a series of .fastq files containing sgRNA data from sorting experiment (Basic protocol 1, 3 and 4 above). Only Read 1 data are needed due the nature of the NGS analysis. **2)** a .csv lookup table containing the names and sequences of the sgRNAs. This file should be formatted with the data in the following columns: i) sgRNA_name, ii) sgRNA_sequence, iii) Gene Targeted

4. Place both the .csv and .fastq files in the same folder (or change command below, as necessary, if at different locations). Run MAGeCK count function on the fastq files (first command in Table 3). Ensure that the conda environment is set to 'mageckenv'.

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### Run MAGeCK test on the count table, in a pairwise way comparing the different samples with the control

5. The second command in Table 3 is used to output sgRNA $\log_2$ fold change statistics and gene-level α-RRA statistics. sgRNA statistics are provided in files with an ".sgrna_summary.txt" suffix, and gene-level α-RRA statistics are provided in files with a ".gene_summary.txt" suffix. The α-RRA statistics files have an aggregated $\log_2$ fold change in them, but the p-values in this file come from the α-RRA analysis.

   The significance of sgRNA enrichment or depletion follows similar methodologies to the RNA-Seq differential expression DESeq2 algorithm. First, the mean and variance of each sgRNA within unsorted and sorted populations is estimated. Moderated variance estimates are computed to reduce technical and biological noise in the data. This is done by fitting a polynomial function between the variance and the mean. Enrichment or depletion of sgRNAs is determined by finding the probability of observing values greater than or equal to, or less than or equal to the mean read sgRNA count in the sorted population, respectively. The probability is evaluated using a Negative Binomial (NB) distribution with the mean and dispersion of the sgRNA counts from the unsorted control population.

   The null hypothesis of α-RRA analysis is that, given a ranked list of sgRNA NB p-values, the distribution of rankings for a gene's set of sgRNAs would be uniformly distributed within this rank list. Finding a majority of a gene's sgRNAs ranked highly in the rank list would cause us to reject this null hypothesis, and suggest that the gene may be impactful. Sample results are presented in Fig. 6A–6B for the case of human P-selectin binding to PSGL-1.

### Log-fold change calculation

6. Log2-fold change (LFC) for sgRNA to determine if they are enriched in the sample with respect to non-target controls in the same sample. Ideally, the non-target control would be scramble sgRNA which are included in the CRISPR library. However, since these are absent in glycoGene CRISPR, the calculations in Fig. 6C uses sgRNA corresponding to enzymes targeting carbohydrate sulfation (all CHSTs) and heparan sulfation (all HS6STs, HS3STs and HS2STs) as non-target controls. Thus, LFC= $\log_2$[sgRNA count for target gene/sgRNA count for non-target gene], in given sample. LFC>0 indicates sgRNA enrichment upon lectin/mAb-based selection. Corresponding target genes play a functional role in lectin/mAb binding. LFC≤0 indicates specific guides that are not enriched as their absolute read counts would be similar to non-target control sgRNA.

## REAGENTS AND SOLUTIONS:

Use Milli-Q, Nanopure or equivalent ultrapure water with a resistivity of 18.2 MΩ cm, unless otherwise noted.

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### DMEM-Complete, 500 ml

In a cell culture hood, add 50 ml of Fetal Bovine Serum (Gibco™, 26140079) and 5 ml each of Antibiotic-Antimycotic (100X; Gibco™ 15240062) and GlutaMAX™ supplement (100X; Gibco™ 35050061) to a 500 ml bottle of DMEM high glucose, pyruvate (Gibco™ 11995065). Mix well and store refrigerated (4°C).

### IMDM-Complete, 1000 ml

In a cell culture hood, add 100 ml of Fetal Bovine Serum (Gibco™, 26140079) and 10 ml each of Antibiotic-Antimycotic (100X; Gibco™ 15240062) and GlutaMAX™ Supplement (100X; Gibco™ 35050061) to a 1000 ml bottle of IMDM (Gibco™ 12440046). Mix well and store refrigerated.

### 1X TE (10 mM Tris, 1 mM EDTA, pH 8.0), 100 ml

Mix 1 ml of 1 M Tris, pH 8.0 (Invitrogen™ AM9856) and 0.2 ml of 0.5 M EDTA (Invitrogen™ AM9261) with 98.8 ml of water. Store at room temperature.

### 50X TAE stock (2M Tris-acetate, 50 mM EDTA), 1000 ml

Weigh 242 g Tris Base (Sigma 93362) and dissolve in 700 ml of water by stirring. Add 57.1 ml of Glacial Acetic Acid (Sigma 695092) and 100 ml of 0.5 M EDTA, pH 8.0 (Invitrogen™ AM9261). Add water to a final volume of 1000 ml. Autoclave (121°C, 15 min) and store at room temperature.

### 2X HBS (50 mM HEPES, 10mM KCl, 12 mM Glucose, 280 mM NaCl, 1.5 mM $Na_2HPO_4$), 100 ml

To 80 ml of water, add 1.192 g HEPES (Sigma H4034), 75 mg KCL (Sigma P5405), 216 mg Glucose (Sigma G7021), 1.636 g NaCl (Sigma, S5886) and 40 mg $Na_2HPO_4.7H_2O$ (Sigma 431478). Stir to dissolve completely. Adjust the pH to 7.05 by dropwise adding 10 M NaOH (Sigma 72068). Make up the volume to 100 ml with water and filter sterilize using 0.2 μm Vacuum filter unit (Thermo Scientific™ 5640020). Store refrigerated.

Correct pH setting is very critical for this reagent to provide good transfection efficiency.

### 1X HEPES buffer (30 mM HEPES, 110 mM NaCl, 10 mM KCl, 10 mM Glucose, 2 mM $MgCl_2$), 1000 ml

Weigh and add, 7.149 g HEPES (Sigma H4034), 6.428 g NaCl (Sigma, S5886), 746 mg KCl (Sigma P5405), 1.8 g Glucose (Sigma G7021) and 407 mg $MgCl_2.6H_2O$ (Sigma M2393) to 900 ml of water. Mix well by stirring. Set the pH to ~7.2–7.3 using 10 M NaOH solution. Filter sterilize using 0.2 μm Vacuum filter unit (Thermo Scientific™ 5670020). Store refrigerated.

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### HEPES/Calcium/HSA (1X HEPES buffer, 1.5 mM CaCl2, 0.1% v/v Human Serum Albumin (HSA))

First prepare 1.5 M Calcium chloride by dissolving 8.324 g of anhydrous $CaCl_2$ (Sigma C5670) in 45 ml of water. Make up the volume to 50 ml. Prepare HEPES/Calcium/HSA by mixing 49.75 ml of 1X HEPES buffer with 50 µl of 1.5 M Calcium Chloride and 200 µl of 25% HSA.

## COMMENTARY

### BACKGROUND INFORMATION:

Glycosylation is a common and diverse post-translational modification (PTM) that decorates most human proteins (Neelamegham & Mahal, 2016; Varki, 2017). The most common types of modification on proteins include O-linked glycosylation at Ser/Thr sites, and N-linked glycosylation at Asparagine residues in the Asn-X-Ser/Thr sequon. Glycans also decorate lipids on the cell surface to form ‘glycolipids’ or ‘glycosphingolipids’. Finally, these carbohydrates can be part of long polymeric/repeating glycosaminoglycan chains. There are other carbohydrate modifications also, but these are the most common types. Knowledge in this field continues to grow with newer post-translational modification being uncovered. Glycans regulate or fine-tune a number of fundamental biological processes like protein folding, pharmacokinetics/clearance rates, host-microbiome/pathogen interactions, cell adhesion and cell signaling. Due to these diverse functions, carbohydrate modifications influence the progression of a variety of human ailments including, but not limited to, inflammation, immunity, viral & bacterial pathologies, and cancer metastasis (Neelamegham & Liu, 2011; Varki, 2017). Understanding how glycans are processed during these various biological processes is thus important for our understanding of a broad range basic science and clinical issues.

Over the years, a number of methods have been developed to dissect or perturb the glycome. These include chemical methods to modify glycan biosynthesis using small molecules that perturb specific biological processes (Marathe et al., 2010; Wang et al., 2018; Wang et al., 2021), techniques to knock-down genes using RNA-interference (Buffone et al., 2013), and methods to edit the glycocalyx focused on genetic targeting (Narimatsu et al., 2021). Among the competing methods to perform genetic manipulations, three methods gained prominence. These were based on Zinc-finger nucleases (ZFNs), transcription activator-like effector nucleases (TALENs) and clustered regulatory interspaced short palindromic repeat (CRISPR)/Cas-based RNA-guided DNA endonuclease (Gaj, Gersbach, & Barbas, 2013). While each of these methods has unique advantages, the use of ZFNs and TALENs was restricted due to high-costs and low gene editing efficiency. Both these systems contain chimeric non-specific nucleases (FokI) coupled with protein modules that bind specific DNA sites. The recognition of the target double stranded DNA sites by the DNA-recognizing module was followed by double-stranded break at proximal sites. Error-prone non-homologous end joining (NHEJ) followed, and this often resulted in nucleotide base insertion-deletion(s) that results in frameshift mutations and loss of protein function. Indeed, ZFN based gene-editing was introduced in the Glycosciences in a manuscript that described the “SimpleCell” technology (Steentoft et al., 2011). This elegant work used ZFNs to

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knockout the catalytic activity of the core-1 Galactosyltransferase (C1GalT1) and thus cells produced truncated O-glycans capped at GalNAcα-O-Set/Thr (±sialic acid). As these glycans were small and easy to identify using high-throughput liquid-chromatography mass spectrometry (LC-MS/MS), the development of the SimpleCell technology resulted in the cataloging of 1000s of O-linked glycosylation sites. The era of TALENs was short-lived, as CRISPR tools came online to demonstrate a more cost-effective and efficient method for precision editing of the mammalian genome (Cong et al., 2013; Mali et al., 2013). In the Glycoscience field the power of this approach was demonstrated by knocking-out the α(2–3)sialyltransferase ST3Gal4 as it abolished sialyl Lewis-X expression on human blood leukocytes, and selectin dependent leukocyte-endothelial cell adhesion interactions (Mondal et al., 2015). Extending this approach, CRISPR-Cas9 reagents were developed and applied to abolish core pathways necessary for the biosynthesis of N-linked, O-linked and glycolipid pathways (Mondal et al., 2016; Stolfa et al., 2016). This helped elucidate the differential role for diverse glycan families in controlling leukocyte-endothelial cell binding. Additional notable developments in this field include the application of CRISPR-Cas9 to organ systems to demonstrate the roles for glycolipids in cell signaling, N-glycans in skin wound healing and notch fucosylation in human tissue formation (Dabelsteen et al., 2020). Glycoscience CRISPR tools also reveal potentially important roles for N- and O-glycosylation in SARS-CoV-2 infection (Yang et al., 2020). Finally, a large family of well-defined CRISPR reagents have been developed for individual glycogenes (Narimatsu et al., 2018). The manuscript accompanying this article presents an even larger assortment of sgRNA that are ideally suited for forward genetic screens (Zhu et al., 2021). Overall, the last few years have led to the streamlining of CRISPR technologies and an ever-growing list of applications in the Glycosciences. The stage is now set for system-wide, multi-level interrogation of glycosylation pathways using this tool (Neelamegham & Mahal, 2016) and their integration with growing glycoinformatics knowledge (Liu & Neelamegham, 2015; Neelamegham & Liu, 2011; York et al., 2020).

## CRITICAL PARAMETERS AND TROUBLESHOOTING:

The critical steps of the protocols have been indicated in notes wherever possible. It is still possible to encounter performance issues. Sometimes they may stem from replacing recommended reagents with items from alternate vendors. In this case, the user may have to refer to the supplier's troubleshooting resources. Results can also be affected if a reagent is compromised by improper storage or handling. The following guide may help in resolving such issues.

### Scale up of the GlycoGene CRISPR library

### No colonies or very few colonies in the plates:

- This may happen if arcing happens during electroporation. Water or oil on cuvette surface or bubbles in the sample may cause arcing. Make sure that there is no condensed water outside the cuvettes before pulsing the sample. Always handle the cuvette with gloved hands. After dispensing sample into the cuvette, tap it gently a few times on the bench to remove any air bubbles from the sample.

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- It is also critical to add SOC recovery medium IMMEDIATELY after the pulse. Any delays will considerably decrease the transformation efficiency. Using prewarmed (at 37°C) SOC recovery medium and LB agar plates are also important for optimal recovery of electroporated cells.
- Compromised storage of electrocompetent cells or expired reagent may also result in poor transformation efficiency. Please make sure to use freshly obtained and properly stored electrocompetent cells to ensure reproducibility of the library from the scale up process.

### Too many colonies in all the plates:

- This could result if the carbenicillin is not added to the plates or old or improperly stored plates are used. If poured ahead of the experiments, ensure a 4°C storage temperature and do not use if more than a month old.

### No band or multiple bands in the PCR reaction:

- Using PCR enzyme master mix and/or conditions other than recommended in the protocol. If changing PCR enzyme, determine the primer annealing temperature and use the cycling conditions recommended for that enzyme mix.

## Preparation of lentivirus particles and transduction

### Transfection efficiency is too low:

- LentiX-293T cells should be passed at about 80–90% confluency. If the cells become overconfluent before passaging, discard and thaw a new vial of cells.
- Low passage cells have better transfection efficiency. Use cells with < 20 passages.
- Plasmid should be of high quality and at a concentration of ≥ 1μg/μl. We use NucleoBond Xtra Midi from Macherey-Nagel. It may help to use a lower volume of water or TE to resuspend the final DNA pellet and also allowing enough time for it to dissolve completely. DNA pellets can be left to dissolve in water or TE overnight at 4°C.
- Cell density at the time of transfection is an important factor. Transfection efficiency goes down if the cells are too dense or too sparse. Cells at 50–60% density result in best transfection rates.
- The pH of 2X HBS is an important factor in forming optimal transfection complexes. The quality also goes down with storage. It may help to test out a new batch of 2X HBS in a small-scale transfection experiment before using it for library transfection. It is also important to use freshly prepared 2X HBS.
- It is important to use properly stored, anhydrous Calcium Chloride to prepare the 2M Calcium Chloride solution. Calcium chloride is highly hygroscopic, deliquescent and exothermic. It is important to notice that the salt used in preparing the reagent is dry, free flowing and when dissolved in water the

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solution heats up. Absence of any of these factors may result in improper molarity of the solution and may lead to poor transfection.

- Aerating the transfection mix by introducing bubbles immediately after addition of 2X HBS helps in forming a more efficient transfection complex in our hands.
- The transfection complex should start to develop cloudiness as incubation progresses for 15–20 min. If turbidity is not obvious, it may be because HBS or calcium chloride solutions are not optimal. Sometimes it may help to incubate the transfection mix at 37°C but if turbidity does not develop in the transfection mix, it is best to remake the HBS and Calcium chloride and repeat the experiment.
- Allow the cells to incubate with transfection mixture for at least 6 hours before changing the growth medium.

### Transduction efficiency is too low

- If the transfection efficiency was good during virus production but virus particles are not very infective it may be due to low virus packaging, lost or damaged virus during collection or transduction at very low MOI for the cell line being transduced.
- Lentivirus production requires presence of the three plasmids in the transfection mix in proper ratio. Make sure that plasmid quantitation is correct and all three plasmids are included in the transfection mix.
- To avoid lost or damaged virus particles, lentivirus containing cell culture supernatant should only be filtered using 0.45 μm filter. Virus particles may be damaged if smaller pore sized filter is used. After ultracentrifugation to collect the virus pellet, supernatant should be immediately and carefully disposed without disturbing the pellet. While resuspending the virus pellet, introduction of bubbles must be minimized. Resuspended virus stock should be stored at −80°C as the virus particles are less stable at higher temperatures.
- Cells lines vary in their susceptibility to lentivirus infection and also virus preparations may vary in their virulent particle count. Therefore, MOI for each virus preparation need to be empirically determined for each cell line of interest by doing the titration experiment as detailed in the protocol.

### Cell sorting for establishing transduced cell lines/CRISPR library

### Cells do not recover after cell sorting (or electroporation)

- It is important to keep the cells on ice while setting up the sorting parameters. Sorted cells should be collected in appropriate vessel that contains the growth medium for that cell line. It is also important to minimize the time between sorting and cell culture. Immediately after sorting, cells should be pelleted, resuspended in growth medium and transferred in a cell culture dish to an incubator.

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- Recovery of some cell lines can be challenging after single cell sorting. It is recommended to sort single cells into 1:1 mixture of fresh and conditioned growth medium.
- The electroporation parameters in this protocol are optimized for HL60 cells. For other cell lines, user will need to optimize the conditions.

### Guide representation is low in the glycoGene CRISPR library cells

- In order to have a high sgRNA representation it is important to transduce at least 500–1000 cells per sgRNA at MOI of 0.1 to 0.4. Additionally, a high cell count needs to be maintained at sorting, electroporation and screening steps to avoid loss of sgRNAs from the library.

### Lectin based selection

### Distinct lectin-binding population is not seen after selection and scale up

- Lectin binding is often dependent on divalent cations. For example, P-selectin binding to PSGL-1 on HL60 depends on the presence of calcium as P-selectin is a C-type (calcium-dependent) lectin. Thus, it is critical that the experiment considers the composition of the lectin binding buffer, and makes modifications accordingly.
- The specificity of lectin binding remains problematic as these have often been discovered from plants using specific carbohydrates for identification. However, the actual specificity may vary with cell types and the glycan binding partners may be quite different. One approach to address this has been to undertake exhaustive glycan array experiments and use computational algorithms to uncover the underlying lectin-binding motifs ((Bojar et al., 2022) and references cited therein). But even here, the orientation/presentation of the glycans is non-physiological, not all glycans are (equally) functional, the repertoire of immobilized carbohydrate structures is finite/limited, and the conclusions drawn are biased by analysis assumptions. Nevertheless, sources in literature may reveal some information on lectin binding specificity. Once this information becomes available, it is possible to confirm that the lectin is functional using either exoglycosidases (e.g. sialidase) to confirm the involvement of specific monosaccharides/motifs in binding or competing lectin binding using related mono- or di-saccharides (e.g. fucose for AAL lectin) (Marathe, Chandrasekaran, Lau, Matta, & Neelamegham, 2008). Thus, it is important to confirm partial information about lectin binding specificity before undertaking cytometry-based selection.
- Many lectins are di-, tri- or tetra-valent and their binding to glycans is related to their binding avidity rather than affinity, i.e. these have weak, monovalent binding interactions that can dissociate from cells. To minimize such dissociation, it is best to perform wash steps to remove unbound lectins just prior to cytometry. Do not keep the cells on ice for more than 10–15 min before sorting.

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- More than one sequential sort may be necessary to enrich a specific population. NGS analysis of intermediate sorts may reveal if enrichment is taking place, rather than just relying on flow cytometry analysis.

### Cell viability after sorting is low or too few cells are sorted

- When preparing cells for functional cytometry sorts, it is best to use buffers made using HEPES as it helps closely mimic cell growth conditions and physiological pH 7.2–7.4, as it has pKa in this range. Compared to PBS (phosphate buffered saline) and Tris, HEPES based buffers are more stable. The addition of glucose in this buffer also provides nutrients to maintain viable cells. Color dyes like phenol red that auto-fluoresce should be avoided during cytometry sorts. These steps will help enhance sort quality and also improve post-sort cell viability.
- When performing sorts, the sorted cell density can be low and this can result in loss of specific populations as cells often rely on neighboring cells for growth nutrients. After sorting, thus, it is best to wash the cells and culture them in conditioned media (described above) to enhance recovery.

## STATISTICAL ANALYSIS:

Not applicable

## UNDERSTANDING RESULTS:

These data are provided in individual figures that accompany each of the Basic Protocols

## TIME CONSIDERATIONS:

Basic Protocol 1: 1 week

Basic Protocol 2: 4 week

Basic Protocol 3: 4 weeks

Basic Protocol 4: 2–3 weeks

Basic Protocol 5: 1–2 hours

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## INTERNET RESOURCES:

Virtual Box for LINUX emulation: www.virtualbox.org

MAGeCK software: https://sourceforge.net/p/mageck/wiki/Home/.

MiniConda for python package installation: https://docs.conda.io/en/latest/miniconda.html

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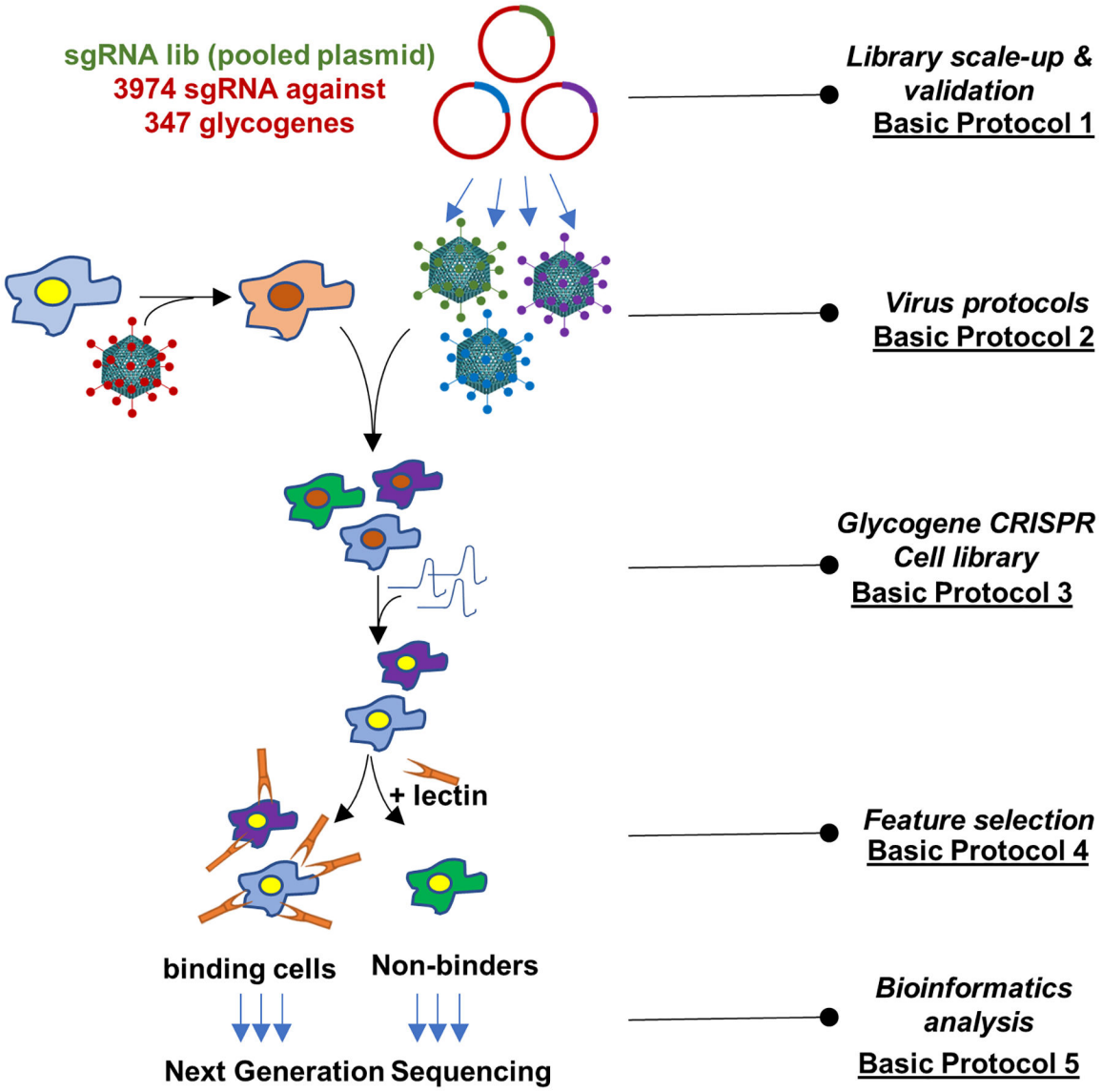



**Figure 1.**
Overall protocol workflow

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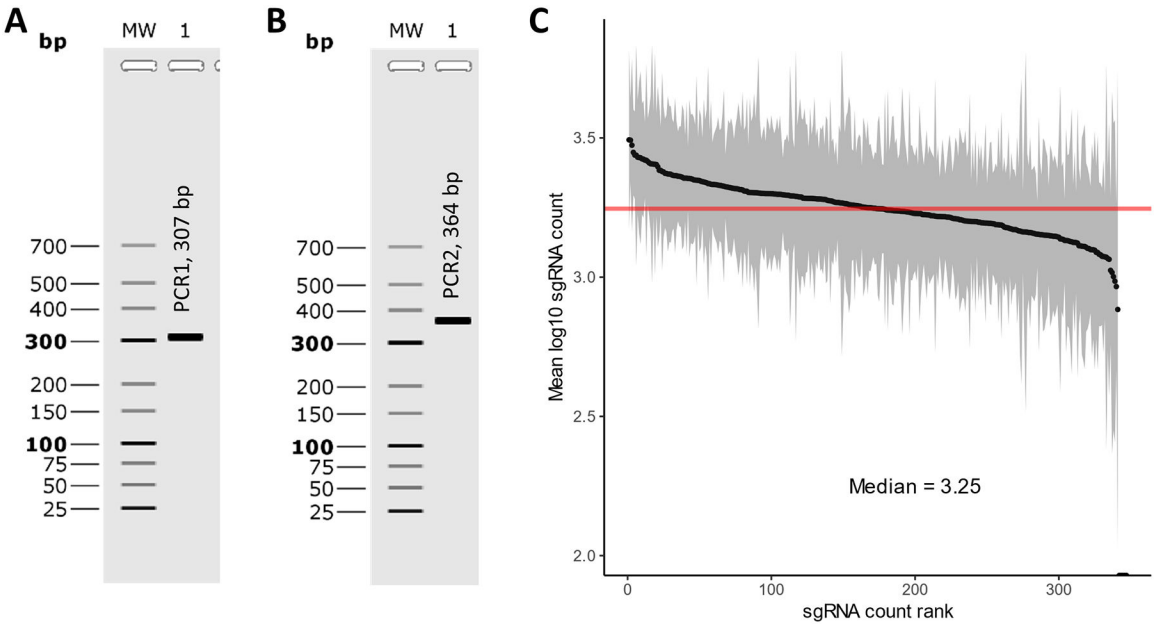



**Figure 2. NGS sample preparation and expected results**

**A**. PCR 1 product of 307bp. **B**. PCR 2 product of 364 bp. **C**. The initial representation of sgRNAs in the glycoGene CRISPR library, where each gene's mean log10 sgRNA read counts is ranked from highest to lowest (black line). The gray bands around each point represents the standard deviation in each gene's log10 sgRNA read count. The median of the mean sgRNA log10 read counts is 3.25, corresponding to roughly 1780 mean reads per sgRNA per gene. While few genes have low mean log10 sgRNA read counts, no genes have the majority of their guides lowly represented (50% below the median of the mean log10 sgRNA counts). The data indicates that all genes in the library have a high representation of sgRNAs and can serve as knocked out guides.

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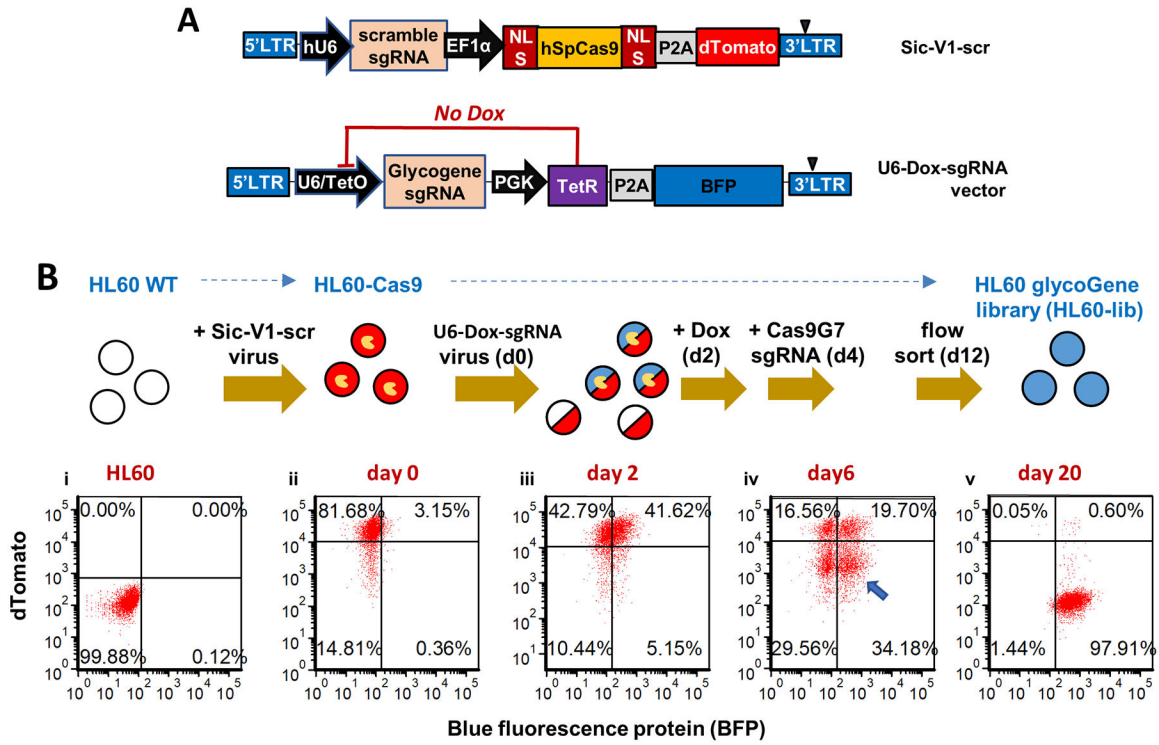



**Figure 3. Creation of glycoGene CRISPR library transduced cells.**

**A**. Schematic of lentiviral vectors used in the CRISPR screen. dTomato acts as a surrogate reporter of Cas9 nuclease activity in SiC-V1-scr. GlycoGene CRISPR library was cloned in a doxycycline-inducible vector with BFP fluorescence reporter. **B**. An HL60-Cas9 cell line was established by transduction with SiC-V1-scr vector, and single cell flow sorting for isogenic clone expressing high levels of dTomato reporter (**Bi–ii**). A virus pool synthesized from the glycoGene CRISPR sgRNA library (U6-Dox-sgRNA) was applied to these cells. Two days later, 42% of the cells were BFP positive (**Biii**). 1μg/mL Dox was added starting at day 2 to initiate gene editing. Dox was removed and Cas9G7 sgRNA electroporated on day 4 to inactivate Cas9 activity. $dTomato^{-}BFP^{+}$ cells (arrow) were FACS sorted at day 12 in order to establish the stable HL-60 glycoGene cell library (called "HL60-lib"). Flow cytometry plots represent: **i**. wild-type HL60; **ii**. HL60-Cas9; **iii**. HL60-Cas9 with glycoGene library; **iv**. cells after Cas9 nuclease editing on day 6; and **v**. HL60-lib cells on day 20. dTomato fluorescence of HL60-lib on day 20 (**Bv**) is similar to wild-type HL60s (**Bi**) confirming complete knockdown of Cas9 activity. (reproduced from Zhu et al. *Glycobiology* **31**:173–180, 2021, with permission).

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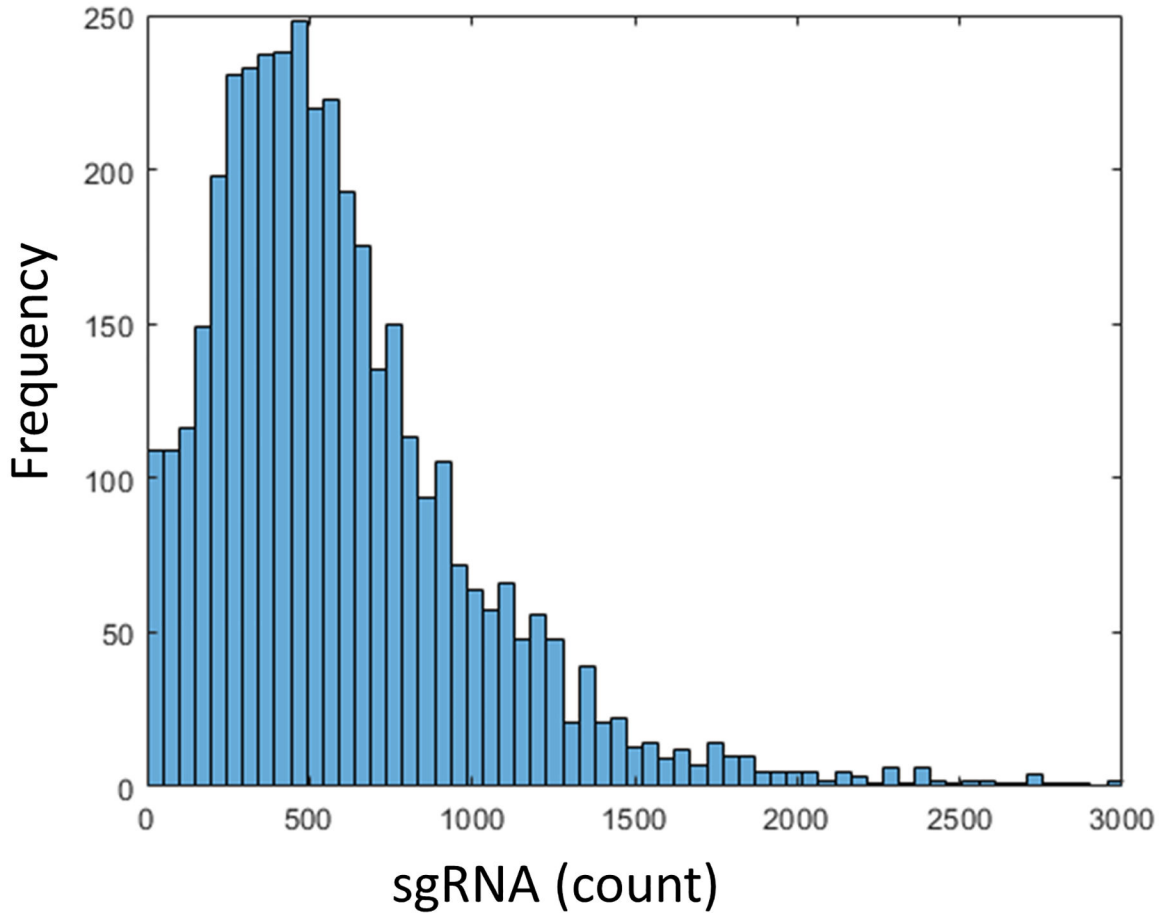



**Figure 4. Guide representation in HL60-lib cells.**

The full glycoGene CRISPR library contains 3974 sgRNA targeting 347 genes. The figure provides the sgRNA distribution in the HL60-lib cells based on Next Generation Sequencing (NGS). None of the glycogenes were lethal for cell growth or appreciably altered cell growth kinetics.

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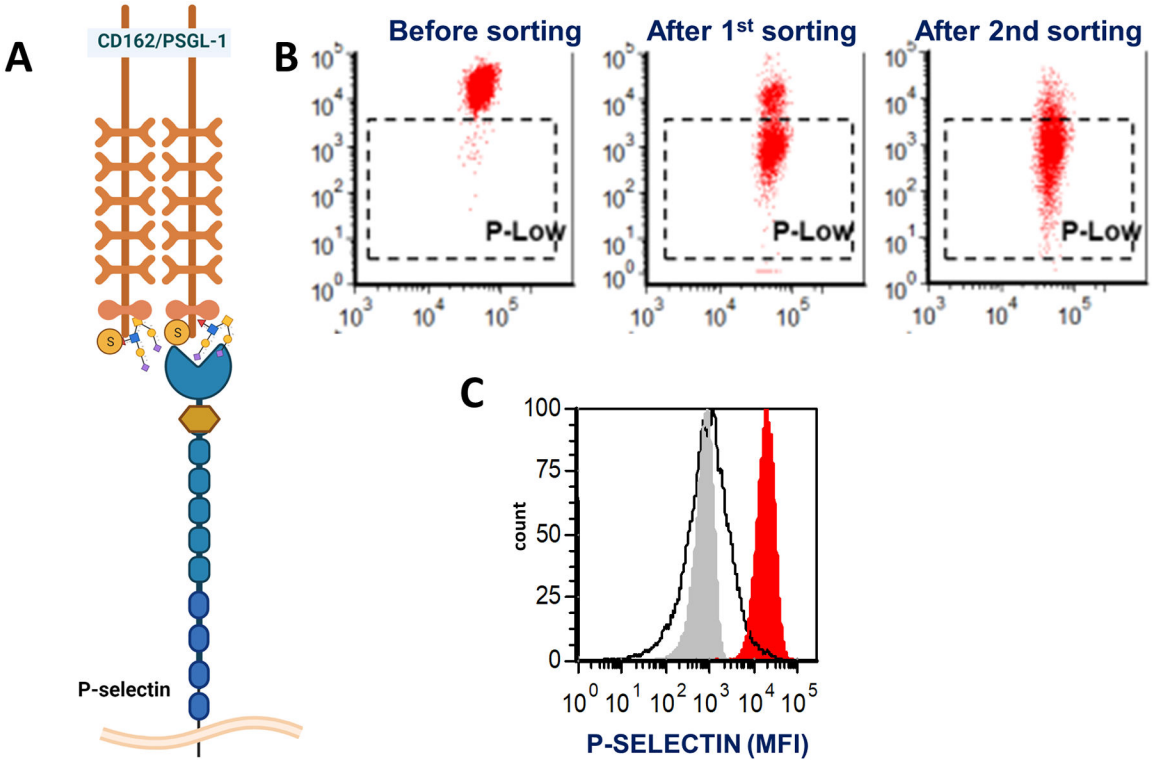



**Figure 5. Flow sorting based negative selection for P-selectin non-binders.**

**A**. P-selectin (CD62P) is expressed at high levels on the surface of stimulated vascular endothelial cells and activated blood platelets. During a variety of human inflammatory and thrombotic diseases, selected carbohydrate epitopes located at the N-terminus of the human blood cell glycoprotein (P-selectin glycoprotein ligand-1. CD162) bind P-selectin with high affinity and strength under shear conditions found in flowing blood. This N-terminal selectin binding epitope contains both a sialyl Lewis-X type carbohydrate epitope, with α(2–3) linked sialic acid and α(1–3) linked fucose, and sulfated tyrosine residues. **B**. In forward genetic screens, two successive sort cycles were used to select a glycoGene CRISPR cell library subpopulation displaying low/no binding to soluble P-selectin. **C**. Histogram shows the P-selectin binding function of the unsorted cells (red), negative control (secondary Ab control) and final P-selectin low selected population (grey line). (Data taken from Zhu et al. *Glycobiology* **31**:173–180, 2021, with permission)

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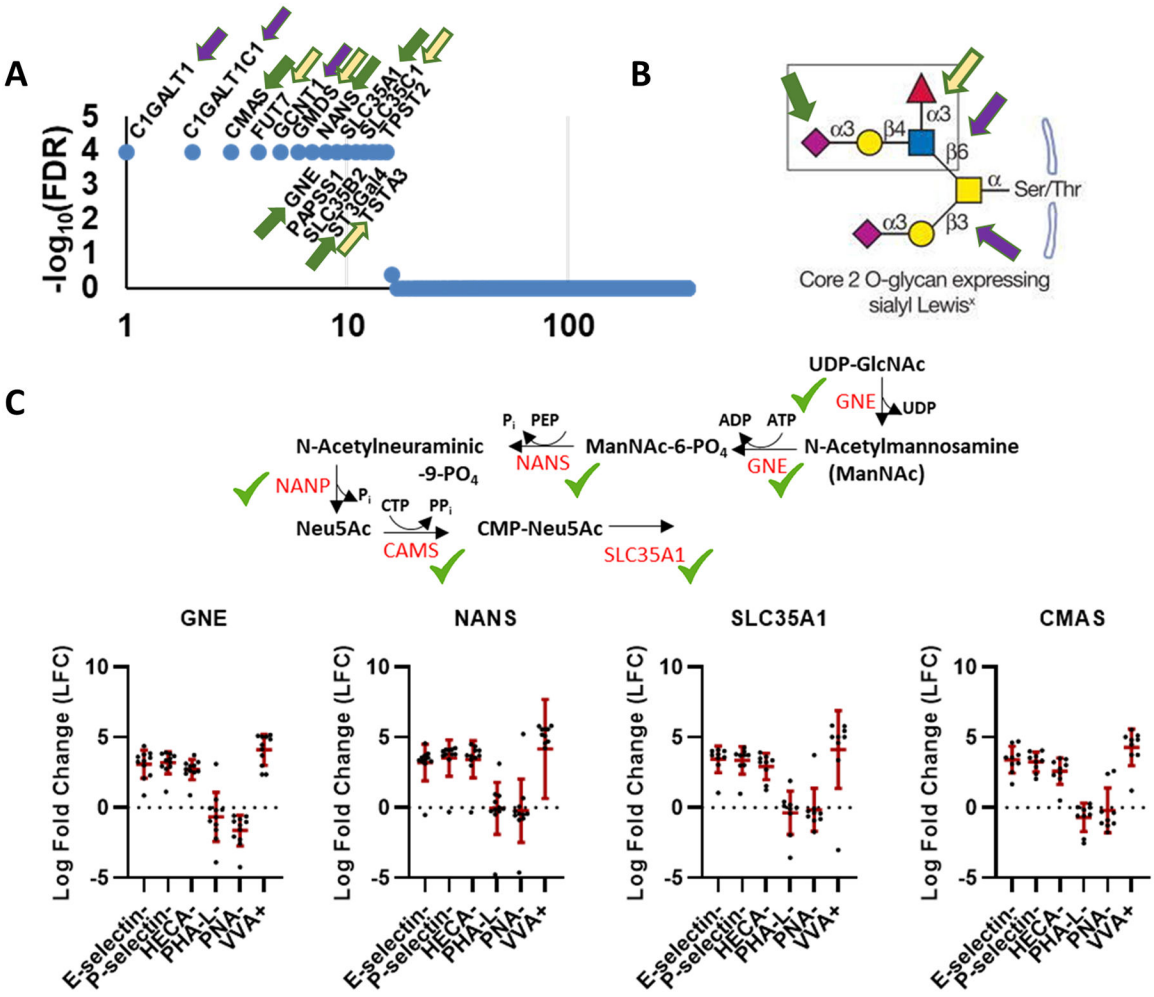



**Figure 6. Determinants of human P-selectin binding to PSGL-1.**

**A**. $Log_{10}$(FDR) calculated using MAGeCK identify several prominent genes regulating P-selectin binding. These genes regulate O-glycan core-1 and core-2 branching (purple arrow), α(2–3)sialylation (green arrow) and α(1–3)fucosylation (yellow arrow). **B**. The exact carbohydrate structures that are modified is indicated using a Symbol Nomenclature For Glycans (SNFG) cartoon. **C**. Log fold-change data are shown for several of the glycoEnzymes regulating sialic acid biosynthesis (i.e. CMP-sialic acid/Neu5Ac formation), with pathway schematic indicating the exact reactions that they catalyze. As seen, all reactions in the CMP-Neu5Ac biosynthesis pathway were identified in the forward genetic, negative-selection screen for P-selectin PSGL-1 binding. These enzymes were also observed to be critical for E-selectin binding and HECA-452 mAb binding. While these genes were not important in negative selection screens for PHA-L and PNA lectin binding, knocking out these gene prominently enhanced VVA-lectin binding. Finally, almost all the sgRNA regulating CMP-Neu5Ac were expressed at high levels in the negative-selection/drop-out screens indicating that a majority of the sgRNA in the glycoGene CRISPR library display high editing efficiency. (Data taken from Zhu et al. *Glycobiology* **31**:173–180, 2021, with permission)

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**Table 1:**

Installation of miniconda in virtualBox environment

| Linux Command | Meaning |
|---|---|
| mach@mach-virtualBox:~$ cd Downloads | change to Downloads/folder where the installation file is miniconda located |
| mach@mach-virtualBox:~$ ls -al | verify that the downloaded file is present |
| mach@mach-virtualBox~/Downloads$ chmod +x Miniconda3-py39_4.10.3-Linux-x86_64.sh | Make the miniconda shell script executable |
| mach@mach-virtualBox~/Downloads$ ./Miniconda3-py39_4.10.3-Linux-x86_64.sh | install miniconda, accept license and all options |
| mach@mach-virtualBox~/Downloads$ source ~/.bashrc | Re-source bash shell configuration to allow “conda” command execution |
| (base)mach@mach-virtualBox~$ conda config –-set auto_activate_base false | Optional: if you do not want conda open by default when you open a terminal |

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**Table 2:**

Installing and updating MAGeCK

| Linux Command | Meaning |
|---|---|
| mach@mach-virtualBox~$ conda activate base | activate the base miniconda environment in a new terminal |
| (base)mach@mach-virtualBox~$ conda config --add channels bioconda | add bioconda channel to minoconda |
| (base)mach@mach-virtualBox~$ conda config –showchannels | confirm that the new channel has been added (you should see bioconda in the terminal output) |
| (base)mach@mach-virtualBox~$ conda create -c bioconda -n mageckenv mageck | create a new environment ‘mageckenv’ and install mageck from bioconda |
| (base)mach@mach-virtualBox~$ conda activate mageckenv | Activate the newly generated MAGeCK environment |
| (mageckenv)mach@mach-virtualBox~$ conda list | confirm that the MAGeCK package is properly installed in the virtual environment |
| (mageckenv)mach@mach-virtualBox~$ conda update mageck | update packages, if necessary to latest version |

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**Table 3:**

Statistical analysis using MAGeCK

| Linux Command | Meaning |
|---|---|
| (mageckenv)mach@mach-virtualBox~$ mageck count -l sgRNA_library.csv -n count_out --norm-method total --sample-label “sample_1,sample_2,control” --fastq fastq_1.fastq fastq_2.fastq fastq_3.fastq | Note: normal-method is “total”. This command will output a set of .tsv file with the prefix “count_out” in the working directory. The main files are: “….count.txt” which has absolute counts, and “….count_normalized” after data normalization. |
| **Arguments**:<br>-l : the sgRNA library file mentioned in step 1<br>-n : prefix of the count table’s name (output file name)<br>--norm-method: allows you to select different normalization types on count data. Choice of methods are: [none,median,total,control]<br>--sample-label: names of the samples that are read in by count. Make sure they are in the same order as you pass the .fastq files containing the reads. There is one sample-label for each fastq file<br>--fastq: a space-delimited list of .fastq files containing the sgRNA reads for different populations in your study.<br>**NOTE:**<br>*“Median normalization” is performed by default in this type of analysis, which assumes the majority of sgRNAs have similar read count distributions before and after selection. Our experiments, however, enrich a subset of the sgRNA from the full library which results in a corresponding depletion of a large number of the sgRNA as a result of selection/screening. This requires the use of “total read count normalization” instead of median normalization.* | |
| (mageckenv)mach@mach-virtualBox~$ mageck test -k count_out.count.txt -t sample_1 -c control -n sample_1_outprefix | Command to run MAGeCK scoring program |
| **Arguments:**<br>-k : count table generated in step 2<br>-t : test column in count table, treated as the “test” group<br>-c : control column in count table. All log2 fold changes will be computed with respect to this column<br>-n : output file prefix<br>*If there are replicates of a particular test condition, provide the replicate names in a comma-delimited list to the “-t” argument in MAGeCK test.* | |